Hilton Paul

Lead Engineer → Associate Architect — AI-Based Applications & Development

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PROFESSIONAL SUMMARY

Lead Engineer with 12+ years architecting and building software and cloud systems across the reinsurance, insurance, and financial domains — already operating at architecture scope, owning system design, technology selection, and cross-team standards, while carrying the Lead Engineer designation. Track record spans microservices and event-driven architecture (.NET Core, Kafka, CDC), cloud architecture on both AWS and Azure (AKS, Cosmos DB, Document AI), and hands-on retrieval-augmented systems — document embeddings, vector/hybrid search, and automated ETL pipelines in Python — the foundational patterns behind RAG. Currently building toward Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103) and targeting an Associate Architect role focused on AI-based application design and development.

ARCHITECTURE OWNERSHIP HIGHLIGHTS

- Owned end-to-end architecture for microservices systems on .NET Core and Azure Kubernetes Service, including service boundaries, event-driven design (Kafka, CDC), and data architecture (SQL Server, MongoDB, Redis, Cosmos DB), across payments, reinsurance, and lending platforms.
- Defined and enforced architectural standards — CQRS, modular monolith decomposition, resilience patterns (retry/exponential backoff), and package/dependency governance — adopted across engineering teams.
- Architected a Python-based document-intelligence pipeline (embeddings, vector/hybrid search, fuzzy matching) — the retrieval architecture underpinning modern RAG systems.
- Directed cloud security architecture and audits across AWS and Azure environments, including key management, WAF, and CDC-based data integration.
- Mentored engineers on system design and architecture principles as part of ongoing technical leadership.

TECHNICAL SKILL SET

Solution & System Architecture	Microservices & event-driven architecture design • Modular monolith decomposition • CQRS • Cloud architecture (AWS & Azure) • Architecture standards & governance • Technology selection
AI / Retrieval & Generative Systems	Document embeddings • Vector & hybrid search • Fuzzy matching • RAG ingestion pipelines (Python ETL) • Azure Document AI / Document Intelligence | Currently building: Microsoft Foundry, Azure OpenAI, agent/tool orchestration, prompt engineering (toward AI-103)
Core Technologies	.NET Core C# Apache Kafka Python
Software Development	ASP.NET Core Web API Microservices architecture RESTful API design Design patterns (Singleton, Factory Method, Abstract Factory) Agile/Scrum CQRS
Database Technologies	SQL Server , postgresql ,MongoDB Redis Azure Cosmos DB
Cloud and DevOps	Azure (AKS, Storage, Document AI) and AWS cloud services Docker and Kubernetes CI/CD pipelines (Azure DevOps)
Message Brokers and Event Streaming	Apache Kafka (including Kafka Streams and Kafka Connect)
Web Technologies	HTML5, CSS3, JavaScript Front-end frameworks (Angular up to v16, React)
Security	OAuth 2.0 and Keycloak Data encryption via Azure Key Vault and HashiCorp Vault Managed identity and cloud security audits
Performance and Monitoring	Application performance tuning Profiling and debugging ELK stack, Prometheus and Grafana, Application Insights
Version Control and Collaboration	Git & TFS Code review processes

Change Data Capture (CDC) and Data Integration	CDC concepts and methodologies Debezium, SQL Server CDC, Kafka Connect sink connectors CDC troubleshooting and monitoring
Soft Skills	Team leadership and mentoring Problem-solving and decision-making
Continuous Learning	Actively studying Microsoft Foundry, Azure OpenAI, and agentic AI solutions for AI-103 Keeping current with .NET and system design Learning Go

CERTIFICATIONS

- AWS Certified Developer Associate (DVA-C01)
- AWS Governance & Compliance
- AWS Well-Architected Framework
- In progress: Microsoft Certified — Azure AI Apps and Agents Developer Associate (AI-103)

PROFESSIONAL EXPERIENCE

Reply (Valorem Reply)

Project: Carnival Cruise — GHESS (Global Health, Environmental & Safety System 2.0)

Senior Consultant 1 | *Jan 2025 – Present*

- Architected the document-intelligence and offline-search design — defining the embeddings/ETL pipeline, vector + fuzzy search approach, and offline-first data sync strategy between ships and shore — the retrieval architecture behind the platform's RAG-style search.
- Directed technical decisions across offshore and onsite teams, working closely with the architecture group to align the safety document management system to enterprise standards.
- Built an automated document-processing ETL pipeline in Python that generates vector embeddings from safety documents and implemented fast vector and text search with fuzzy matching over a 1GB+ offline document corpus.
- Designed and developed a React web application and web APIs, with offline database sync via RxDB and Kafka-based ship-to-shore data synchronization.
- Handled SQL Server database migrations using Entity Framework.
- Collaborated with international cross-country teams across Europe, the USA, Brazil, and Italy.

NEST DIGITAL

Project: National Payment System (NPSS), UAE

Lead Engineer | *Aug 2023 – Jan 2025*

- Led architecture for a national-scale microservices platform on .NET Core and Azure Kubernetes Service, making the core technology and service-boundary decisions and selecting Azure Document AI and Azure Storage for document-intelligence and storage needs.
- Defined the system's data architecture — CQRS for read/write separation, Redis for distributed caching, and MongoDB (with GridFS) alongside Azure Cosmos DB — to meet scalability and flexibility requirements.
- Owned security architecture and CDC strategy, directing security audits and setting up CDC with Debezium to capture SQL Server changes.
- Established engineering standards through internal NuGet packages and a private NuGet server for secure, reusable distribution across teams.
- Mentored junior developers in system architecture principles and product design methodologies.
- Developed and maintained RESTful APIs using ASP.NET Core Web API and Entity Framework, and long-running background services with a resilient retry mechanism (exponential backoff) for transient failures.

COGNIZANT

Project: AXA XL Reinsurance

Senior Associate — led development and maintenance for one of the world's largest P&C reinsurers using microservices architecture and Azure Functions | *Aug 2021 – Jun 2023*

- Architected large-scale Single Page Applications (Angular 14) for reinsurer services, defining the module structure, lazy-loading strategy, and component/reusability standards adopted across the frontend codebase.

- Drove performance-architecture decisions — OnPush change detection, virtual scrolling, and Infragistics igx-grid virtualization — to support data-heavy views at scale.
- Developed enterprise-level web applications using Web API, C#, and Angular, and implemented automated deployment pipelines using Azure DevOps.

Muthoot Microfin

Project: In-house LMS & LOS

Senior Manager (Technical Lead & Cloud Audits) | *Jul 2018 – Aug 2021*

- Architected and led the migration of a legacy .NET 3.5 monolith to a modular monolithic .NET Core 3.1 (later .NET 5) service on AWS, defining the module decomposition strategy while preserving a single deployment unit.
- Designed the target AWS cloud architecture (EC2, RDS for SQL Server, CloudWatch) and directed security architecture, including regular audits and AWS WAF.
- Refactored legacy code to SOLID principles with dependency injection and inversion of control throughout the application.
- Implemented responsive Angular Material UI and reusable Angular services for backend communication.
- Improved application performance by 60%, successfully modernizing a 10-year-old monolith into a scalable AWS-based architecture.

SysGsoft Solutions

Project: Ecommerce and AI Chatbot

Technical Lead | *Jan 2018 – Jul 2018*

- Developed and maintained an ecommerce system with Web API, Angular frontend, and Windows Services for background processing.

Technowave Solutions

Project: Logistics

Software Developer | *Mar 2017 – Jan 2018*

- Developed and maintained a logistics system with Web API, Angular frontend, and Windows Services.

Rainybits Technology

Project: Sedaru Water Utility Management Software

Software Developer | *Sep 2013 – Mar 2017*

- Developed and maintained web applications for the utility and water management sectors using AngularJS and RESTful Web APIs.

EDUCATION

- Master of Computer Applications (MCA) — CMS College of Science and Commerce, Bharathiar University (2010–2013)
- Bachelor of Science in Computer Science — CMS College of Science and Commerce, Bharathiar University (2007–2010)

DECLARATION

I, Hilton Paul, hereby declare that all the information provided in this resume is true and accurate to the best of my knowledge.